FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

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THE PATENTS RULES, 2003

COMPLETE SPECIFICATION

(See section 10 and rule 13)

1. Title of the Invention

IoT enabled robot for detecting gas pipe leakage

2. Applicant(s)

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3. Preamble to the description

The following specification particularly describes the invention and the manner in which it is to be performed.

IoT enabled robot for detecting gas pipe leakage

FIELD OF INVENTION

The present invention relates to the industrial gas pipe leakage detection and more particularly it relates to an IoT enabled robot for detecting gas pipe leakage.

BACKGROUND OF INVENTION

CN114414571B describes a wireless robot is made to detect toxic gases, especially in tight or hard-to-reach spaces. It consists of a main body, a set of rolling tracks (like tank treads), and a part that holds the gas detector. The rolling tracks are mounted on the sides of the robot, and there are at least two of them to help it move around. The gas detector can rotate, allowing it to adjust its position for better detection. Inside the robot's body, there is a driving mechanism that powers the tracks using an electric motor and a system of gears and rods. This mechanism allows the robot to move easily, even in narrow areas. The robot does not need to rely on wires, making it more convenient and safer to use in tight spaces. Overall, this design improves the control and efficiency of toxic gas detection in confined areas.

ES2901649T3 comprises an inspection robot consisting of a plurality of sleds where each sled can carry a sensor to measure the characteristics near the surface being traversed, such as emission sensors to test gas leaks, air quality monitoring, radioactivity, the presence of liquids, electromagnetic interference, visual data of the surface being traversed, such as uniformity, reflectance, condition of coatings, such as epoxy coatings, wall thickness values or patterns, wear patterns, and the like.

202441036414 features a network of IoT-enabled gas sensors strategically placed near pipelines, storage tanks, and gas-processing equipment to monitor gas concentrations and environmental conditions such as temperature and humidity. The data collected by these sensors is transmitted wirelessly to a central processing

unit, where it undergoes preprocessing to remove noise and enhance data quality. The preprocessed data is then analyzed by a CNN model, trained to identify patterns associated with gas leaks, enabling accurate and efficient detection while minimizing false positives. The IoT-CNN system provides real-time alerts when potential gas leaks are detected, allowing for immediate action through various channels such as SMS, email, or integration with industrial control systems. The system's continuous monitoring and pattern recognition capabilities offer several advantages over traditional gas leak detection methods, including higher accuracy, reduced false alarms, and faster response times. It also supports remote monitoring, scalability, and predictive maintenance strategies, which contribute to enhanced industrial safety and environmental sustainability.

202441036304 is a gas pipe leakage detection system that uses an Arduino Nano and MQ4 gas sensor to address this risk. It includes an LCD display for real-time gas concentration monitoring and a buzzer for audible alerts when gas levels exceed a threshold. At the heart of this system are the sensors, meticulously placed along the pipeline network. The sensors immediately transmit alerts to the control unit via wireless communication protocols. The control unit serves as the central hub of the system, receiving and processing data from the sensors in real-time.

202411024301 consists of gas sensors that are strategically positioned in the industrial facility to monitor air for hazardous gases. A microcontroller is used to process the sensors data and a cloud platform manages the data from multiple sensors. Processed data is transmitted to a central server or cloud platform via WIFI and GSM through user-friendly web or mobile interfaces. The system continuously monitors gas levels and triggers alerts for abnormal concentrations or leaks, and notifying personnel via SMS or push notifications.

202411026921 employs an automated ignition circuit to ensure continuous operation and safety. When the flame thermostat sensor detects no flame, it ignites the gas stove burner preventing combustible gas accumulation.

202441030458 is a system for gas leak detection and threat zone identification. It includes multiple gas sensors that transmit real-time air quality and data of harmful

gases to computing devices. These devices then analyze the data, considering the environmental factors like wind speed, direction and temperature, to detect for gas leakages and identify threat zones of varying severity.

202431027449 describes an AI based LPG gas leakage detection system and fire alert system notification. The invention is centered around the Arduino UNO R3 microcontroller. The system includes a Gas Sensor (MQ2) connected to the Arduino to detect gas leaks, which triggers a buzzer for immediate alert. To prevent ignition sources, a relay is incorporated to cut off the electricity, and a servo motor is used to turn off the gas knob, minimizing the risk of leakage. These components work together to provide a robust gas safety mechanism within the home. In addition to gas leak detection, fire detection is managed through a NodeMCU ESP8266 module connected to a fire sensor. When the sensor detects fire, it sends a signal over the internet to an AI system, developed in Python, for analysis. If the fire is confirmed, the AI system automatically sends a WhatsApp message to notify the homeowner of the emergency. This integration of IoT devices and AI enhances safety by enabling real-time alerts, ensuring prompt action to mitigate gas and fire hazards while communicating effectively with homeowners.

202441024571 comprises of gas sensors that are connected to a central control unit which analyzes data and triggers safety measures. It also includes a solenoid valve to shut off gas supply upon detection of a leak, and a ventilation module to disperse the gas fumes for enhanced safety.

202431022160 consists of a load cell that monitors LPG gas weight and transmits values to a microcontroller. If gas levels fall below a set threshold, a notification is sent to the gas supplier for cylinder replacement along with a status update to the customer. IoT addresses safety concerns, ensuring timely alerts for gas leakages.

202441017128 employs strategically positioned sensors to detect methane, propane, and carbon monoxide in real-time. An AI algorithm is used to analyze data to identify and assess gas leaks, which smart gas meters are used to monitor cylinder levels. In case of gas leaks alerts are sent through various channels enhancing safety.

GB2544529A comprises moving at least one robot or robotic system as defined above along a pipeline within a pipeline network; inspecting said pipeline for leaks or failures using the at least one machine vison system and/or the at least one NDT device and/or the at least one other sensor device of said robot or robotic system; and tracking the position of said robot or robotic system within said pipeline using the at least one locating device, e.g. a global positioning system (GPS).

However, there does not exist a system that involves an IoT enabled robot that moves on top of gas pipes providing 360-degree rotation to remotely monitor and detect LPG leakages in real-time even in low-light conditions with image capture.

OBJECTIVE OF THE INVENTION

- The primary objective of the invention is to provide a cost-effective solution for real-time monitoring and detection of industrial gas pipe leakages.
- Another objective of the present invention is to make a robot that can navigate through confined or hazardous spaces where human access is limited or risky.

SUMMARY

The present invention is described hereinafter by various embodiments. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiment set forth herein.

According to the first aspect of the present invention, there is provided an IoT enabled robot for detecting gas pipe leakage. The IoT enabled robot for detecting gas pipe leakage comprises a processing unit, an ultrasonic sensor, motors, a motor driver, an imaging device, a torch, a gas sensor, a temperature and humidity sensor, a castor wheel, wheels, a data store and a monitoring and control center.

Further, the processing module is made to receive data from the gas sensor, temperature and humidity sensor to detect for gas pipe leakages.

In accordance with an embodiment of the present invention, the processing unit tracks the location of gas leakage and sends this information to the monitoring and control center.

In accordance with an embodiment of the present invention, processing unit sends alerts to the monitoring and control center in the event of a gas leak.

In accordance with an embodiment of the present invention, the ultrasonic sensor measures the distance to obstacles, preventing the robot from colliding with them.

In accordance with an embodiment of the present invention, the monitoring and control center can instruct the processing unit to capture images using the imaging device and provide illumination with the torch in low-light conditions.

BRIEF DESCRIPTION OF THE DRAWINGS

So that the manner in which the above recited features of the present invention can be understood in detail, a more particular description of the invention, briefly summarized above, may have been referred to by embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this invention and are therefore not to be considered limiting of its scope, for the invention may admit to other equally effective embodiments. These and other features, benefits, and advantages of the present invention will become apparent by reference to the following text figure, with like reference numbers referring to like structures across the views, wherein:

FIG.1 is a perspective view of an IoT enabled robot for detecting gas pipe leakage in accordance with an embodiment of the present invention;

FIG.2 is a top view of an IoT enabled robot for detecting gas pipe leakage in accordance with an embodiment of the present invention;

FIG.3 is a frontal view of an IoT enabled robot for detecting gas pipe leakage in accordance with an embodiment of the present invention;

FIG.4 is a right-side view of an IoT enabled robot for detecting gas pipe leakage in accordance with an embodiment of the present invention;

FIG.5 illustrates the communication process of the invention, an IoT enabled robot for detecting gas pipe leakage in accordance with an embodiment of the present invention;

LIST OF COMPONENTS

1 - Processing Unit
2 - Ultrasonic Sensor
3 - Motors
4 - Motor Driver
5 - Imaging Device
6 - Torch
7 - Gas Sensor
8 - Temperature and Humidity Sensor
9 - Castor Wheel
10 - Wheels
11 - Monitoring and Control Center
12 - Data Store

DETAILED DESCRIPTION OF THE DRAWINGS

Figure 1. is the perspective view of an IoT enabled robot for detecting gas pipe leakage. The IoT enabled robot continuously monitors the gas sensor (7) and temperature and humidity sensor (8) readings to detect any gas pipe leakages. A processing unit (1) sends the data of these sensors to a data store (12) which is then displayed to a monitoring and control center (11). An ultrasonic sensor (2) prevents the robot from colliding into any obstacle by measuring the distance between the robot and the obstacle using ultrasonic sound waves. A torch (6) provide illumination in low-light conditions and an imaging device (5) is used to provide visual inspection. A motor driver (4) is used to rotate the wheels (10) through motors (3).

Figure 2. is the top view of an IoT enabled robot for detecting gas pipe leakage. It illustrates the positioning of the processing unit (1), the gas sensor (7), the

temperature and humidity sensor (8), the torch (6), the castor wheel (9), the plurality of wheels (10), the ultrasonic sensor (2), the imaging device (5), and the motor driver (4).

Figure 3. is the frontal view of an IoT enabled robot for detecting gas pipe leakage. This figure illustrates the positioning of an imaging device (5), an ultrasonic sensor (2), a gas sensor (7), a temperature and humidity sensor (8), a castor wheel (9), wheels (10) and a torch (6).

Figure 4. is the right-side view of an IoT enabled robot for detecting gas pipe leakage. This figure illustrates the positioning of a processing unit (1), an imaging device (5), a temperature and humidity sensor (8), a torch (6), a motor (3), a wheel (10), and a castor wheel (9).

Figure 5. is the top view of the invention, an IoT enabled robot for detecting gas pipe leakage which illustrates the communication process of the invention. This figure illustrates the positioning of the robot on gas pipe and how it interacts with the monitoring and control center (11) through the data store (12).

DETAILED DESCRIPTION OF THE INVENTION

The present invention is described hereinafter by various embodiments with reference to the accompanying drawing, wherein reference numerals used in the accompanying drawing correspond to the like elements throughout the description.

While the present invention is described herein by way of example using embodiments and illustrative drawings, those skilled in the art will recognize that the invention is not limited to the embodiments of drawing or drawings described and are not intended to represent the scale of the various components. Further, some components that may form a part of the invention may not be illustrated in certain figures, for ease of illustration, and such omissions do not limit the embodiments outlined in any way. It should be understood that the drawings and detailed description thereto are not intended to limit the invention to the particular form disclosed, but on the contrary, the invention is to cover all modifications,

equivalents, and alternatives falling within the scope of the present invention as defined by the appended claim.

The present invention is an IoT enabled robot for detecting gas pipe leakage that has a processing unit (1), a gas sensor (7), a temperature and humidity sensor (8), an ultrasonic sensor (2), motors (3), a motor driver (4), an imaging device (5), a torch (6), a castor wheel (9), wheels (10) as seen in figure.1. and a monitoring and control center (11) and a data store (12) as seen in figure.4.

The IoT enabled robot for detecting gas pipe leakage is made to address the critical need for real-time gas pipe leakage detection and remote monitoring using IoT technology. At its core, the project incorporates a processing unit (1) orchestrating the intricate functionalities of the robot.

The robot is made to move on top of industrial gas pipes to detect if there is a leakage as seen in figure.4.

The best method of performing the invention, an IoT enabled robot for detecting gas pipe leakage is disclosed as follows:

The robot connects to the Internet and is positioned on top of industrial gas pipes to begin leak detection. A monitoring and control center (11) manages the robot's movements, including left, right, forward, and backward directions. It sends a command to a data store (12), which then transmits a signal to a processing unit (1). Upon receiving the signal, the processing unit (1) forwards it to a motor driver (4), which directs the motors (3) to rotate in a specific direction. As the motors (3) rotate, the attached wheels (10) turn, causing the robot to move accordingly. The presence of a castor wheel (9) allows for 360-degree movement, improving the robot's mobility.

An ultrasonic sensor (2) positioned at the front continuously measures the distance to any obstacles that could hinder the robot's movement using ultrasonic sound waves. When an obstacle is detected, the ultrasonic sensor (2) sends a signal to the processing unit (1), which then instructs the motor driver (4) to stop the motors (3). Consequently, the motors (3) cease rotating, causing the wheels (10) to stop, thereby halting the robot, and preventing a collision with the obstacle.

The robot is equipped with a temperature and humidity sensor (8) as well as a gas sensor (7), both crucial for detecting gas leaks. The temperature and humidity sensor (8) continuously monitors the surrounding conditions and sends this data to the processing unit (1), which then forwards it to the data store (12). This information is subsequently retrieved by the monitoring and control center (11). At the same time, the gas sensor (7) continuously checks for the presence of gas leaks and sends the data to the processing unit (1) for further processing.

Data from both the temperature and humidity sensor (8) and the gas sensor (7) is transmitted from the processing unit (1) to a machine learning (ML) model deployed on the data store (12). The ML model analyzes parameters such as temperature, humidity, and gas levels to determine the likelihood of a gas leak. The gas leakage status is then stored in the data store (12) and subsequently updated to the monitoring and control center (11). By incorporating multiple parameters like temperature, humidity, and gas levels, the model enhances its accuracy in detecting gas leaks.

In addition to its sensing capabilities, the robot is equipped with an imaging device (5) that enables remote visual inspection through image capture and viewing. The monitoring and control center (11) communicates with the imaging device (5) via the processing unit (1) to capture images, which are then stored in the data store (12). These images are later retrieved to help assess the robot's surroundings and potentially identify the source of any gas leaks. This feature adds versatility, allowing users to visually evaluate the situation and make more informed decisions.

Additionally, the robot is equipped with a torch (6) to provide illumination in low-light conditions, ensuring visibility and functionality in challenging environments. The torch (6) can be activated remotely by the monitoring and control center (11). It provides essential lighting in dim settings, assisting with visual inspections and enabling the capture of clear images in darkness, which would not be possible without the torch (6).

The robot's location is continuously tracked and relayed to the monitoring and control center (11). This real-time tracking is valuable for monitoring the robot's movements and enabling swift action at the site in the event of a detected gas leak.

Finally, the robot has the capability to send immediate alerts in the event of gas leaks to the monitoring and control center (11), ensuring that they are promptly informed and can take prompt action.

The seamless integration of the data store (12) with the monitoring and control center (11) and the processing unit (1) facilitates efficient data exchange and synchronization. This cloud-based architecture allows for real-time updates, remote management, and data storage, significantly enhancing the project's scalability and responsiveness.

In summary, the IoT-enabled robot for detecting gas pipe leaks represents an integration of hardware, software, and cloud technologies is made for the early detection of leaks in industrial gas pipes. Utilizing a temperature and humidity sensor (8), a gas sensor (7), and a machine learning algorithm, the robot effectively monitors for leaks and promptly sends alerts to the monitoring and control center (11). The ultrasonic sensor (2) helps prevent collisions with obstacles, thereby avoiding damage. Additionally, the imaging device (5) and torch (6) enhance the robot's functionality by capturing images of its surroundings, even in low-light conditions. The robot's location tracking capabilities provide valuable information about its position, enabling appropriate responses in the event of gas leaks. Its functionalities extend beyond detection, incorporating real-time monitoring, remote management, and visual inspection. This comprehensive approach to gas leak detection and safety highlights the potential of IoT applications in addressing critical environmental and safety challenges in various industrial sectors.

We claim:

1. An IoT enabled robot for detecting gas pipe leakage (100) comprising:
   a processing unit (1);
   an ultrasonic sensor (2);
   a motor driver (4);
   plurality of motors (3) attached to their respective wheel (10);
   an imaging device (5);
   a torch (6);
   a gas sensor (7);
   a temperature and humidity sensor (8);
   a castor wheel (9);
   a monitoring and control center (11);
   a data store (12);
   characterized in that

   an IoT enabled robot for detecting gas pipe leakage (100) navigates through spaces controlled by the monitoring and control center (11);

   wherein the ultrasonic sensor (2), the motor driver (4), the imaging device (5), the torch (6), the gas sensor (7), and the temperature and humidity sensor (8) are connected to the processing unit (1);

   wherein the processing unit (1) accepts the data from the ultrasonic sensor (2), the gas sensor (7), the temperature and humidity sensor (8), and the imaging device (5);

   wherein the processing unit (1) sends the data to the monitoring and control center (11);

   wherein the processing unit (1) processes the accepted data to detect gas leakage;

   wherein the processing unit (1) triggers alerts when gas leakage is detected;

2. The IoT enabled robot for detecting gas pipe leakage (100) as claimed in claim 1, wherein the processing unit (1) communicates the location of IoT enabled robot for detecting gas pipe leakage (100), in presence of gas leakage.

3. The IoT enabled robot for detecting gas pipe leakage (100) as claimed in claim 1, wherein the ultrasonic sensor (2) senses the obstacle and communicates the presence of the obstacle to the monitoring and control center (11).

4. The IoT enabled robot for detecting gas pipe leakage (100) as claimed in claim 1, wherein the castor wheel (9) rotates IoT enabled robot for detecting gas pipe leakage (100) within range of 0 - 360 degree.

5. The IoT enabled robot for detecting gas pipe leakage (100) as claimed in claim 1, wherein the monitoring and control center (11) communicates to imaging device (5) through processing unit (1) to capture and send the view to monitoring and control center (11).

Dated this the 15th Day of October 2024

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